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Hadrons and Other Secondaries Generated by Cosmic-Ray Muons Underground.

F. F. KHALCHUKOV(1), E. V. KOROLKOVA(1), V. A. KUDRYAVTSEV(1)
A. S. MALGUIN(1), V. G. RYASNY(1), O. G. RYAZHSKAYA(1)
G. T. ZATSEPIN(1) and O. SAAVEDRA(2)

(1) *Institute for Nuclear Research of the Russian Academy of Sciences - Moscow, Russia*
(2) *Istituto di Cosmogeofisica del CNR - Torino, Italy*
Istituto di Fisica Generale, Università di Torino - Torino, Italy
INFN, Sezione di Torino - Torino, Italy

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Summary. — The production of nuclear-active and electromagnetic particles, as well as long-lived isotopes in nuclear and electromagnetic showers generated by cosmic-ray muons at different depths underground is discussed. Characteristics of the secondary components and their dependence on the depth and the muon energy, obtained in experiments and calculations, are presented.

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1. – Introduction.

Ten years ago the First Symposium on Underground Physics (UP'85) was organized at Saint-Vincent by professor C. Castagnoli. It was the time of intense development of this new branch in physics. It was the time of the beginning of the LSD experiment in the Mont Blanc Laboratory, of the inauguration of the Underground Frejus Laboratory, of the completion of excavation of the Gran Sasso Underground Laboratory. Underground physics includes experiments on elementary-particle and neutrino physics, cosmic-ray physics and astrophysics. The study of cosmic-ray penetrating component is one of the branches of the underground physics. Among many other questions it investigates the problem of the origin of hadrons and other secondaries at various depths.

Research in this direction has started in the '50s, but at present it becomes significant and requires reassessment of many ideas, on which various experiments studying rare processes deep underground have been based. Such experiments can be carried out only in low cosmic-ray background conditions, therefore the detectors have to be placed deep underground, where cosmic rays are represented mainly by muons and neutrinos. While the cosmic-ray muon component can be rejected by

anticoincidence systems, it is more difficult to get rid of nuclear-active secondaries generated by the muons.

In passing through rock, muons generate γ-quanta, electrons, neutrons, protons, pions and long-lived radioactive isotopes mainly in nuclear and electromagnetic showers. These particles, as well as β's and γ's emitted by the muon-produced isotopes, are the background for all experiments deep underground. The possibility of a rare-process detection depends on the ability of suppressing the background from muons and their secondaries. So the knowledge of the secondary cosmic-ray component underground becomes very important.

2. – Hadron production underground.

Nuclear-active particles are produced mainly in nuclear and electromagnetic showers: in the first case they are generated in pion-nucleus and nucleon-nucleus interactions, in the second case in photonuclear reactions.

If a high energy is transferred to the inelastic interaction of a muon, multiple production of hadrons ($\pi^{\pm}$, π^0, $K^{\pm}$, K^0, n and p) occurs, and a nuclear shower develops. The hadrons in the nuclear shower can be divided according to their energy into three groups: shower hadrons ($\beta > 0.7$), cascade hadrons ($\beta = 0.2$–0.7) and evaporative nucleons ($\beta < 0.2$). Shower pions $\pi^{\pm}$ produce new generations of particles until particles with $\beta > 0.7$ no longer appear. The cascade hadrons in collisions with nuclei produce cascade particles with lower energies and evaporative nucleons. The average multiplicity of cascade and evaporative particles per collision is practically constant for incident-particle energies $E > 20$ GeV. The shower particles move mainly in the direction of the primary hadrons. Cascade nucleons and pions have quite a broad angular distribution, but still are sensitive to the direction of the primary particle. Evaporative nucleons are emitted isotropically.

Cascade neutrons are capable of further multiplication in nuclear reactions, while cascade protons rapidly lose their energy by ionization. An additional mechanism of neutron production is capture of slow π^- by a nucleus. Slow π^+'s decay in a chain $\pi^+ \to \mu^+ \to e^+$. The number of slow $\pi^{\pm}$, which are cascade particles, depends on the number of interactions of shower particles with nuclei. The total number $N_{p,n,\pi}$ of both slow $\pi^{\pm}$ and nucleons is directly proportional to the number of interactions of shower particles N_s. As $N_s \propto \overline{E}_s^{0.7}$ (E_s is the shower energy), then $N_{p,n,\pi} \propto \overline{E}_s^{0.7}$. This dependence was first obtained in [1].

The main process as the result of which neutrons and pions are generated in electromagnetic showers is photoproduction in nuclei of matter. Evaporative and cascade particles are mainly produced in photonuclear reactions. The main contribution to the cross-section for interactions of γ's with production of neutrons is from the (γ, n) reaction, which has a giant resonance. In addition to this, it is necessary to take into account multiple photoproduction reactions (γ, xn). As a result of the fact that the spectrum of shower photons falls down with energy as dE_γ/E_γ^2 [2], neutrons in electromagnetic showers are generated mainly by low-energy photons.

However, there are channels of neutron production by γ's of higher energies: first, due to inelastic charge exchange reactions

$$\gamma + p \to n + \pi^+ , \tag{1}$$

$$\gamma + (Z, A) \to (Z - 1, A - 1) + n + \pi^+ , \tag{2}$$

and second, in two-step reactions such as

$$\gamma + (Z, A) \to (Z, A) + \pi^+ \pi^- , \tag{3}$$

$$\pi^- + (Z, A) \to (Z - 1, A - 1) + \mathrm{n} . \tag{4}$$

Pions are produced in photoproduction reactions ($\gamma, x\pi^+ \pi^-$) and also in charge exchange reactions $\gamma + \mathrm{p} \to \mathrm{n} + \pi^+$, $\gamma + \mathrm{n} \to \mathrm{p} + \pi^-$.

In slow-π^- capturing by a nucleus, neutrons are produced. This effect leads to an increase of the multiplicity of neutrons.

The dependence of the average multiplicity of neutrons and pions in electromagnetic showers on the shower energy E_s is satisfactorily approximated by a linear function [1] $\overline{N}^{\mathrm{e.m.}}_{\mathrm{n},\pi} = AE_s^{1.0}$. If we take into account that in the interaction of a high-energy shower photon with matter high-energy pions can be produced, which are capable of producing nuclear showers of low strength, in this case the dependence becomes $\overline{N}^{\mathrm{e.m.}}_{\mathrm{n},\pi} = AE_s^{1.05}$.

3. – Experimental study of hadrons underground.

Underground studies of hadrons were carried out using different scintillation counters [3-5]. The dependence of fluxes of hadrons (neutrons and pions) produced by cosmic-ray muons on the cascade energy and the average muon energy was obtained with the scintillation detector ASD at the Arteomovsk Scientific Station of the Institute for Nuclear Research of the USSR Academy of Sciences. The detector has 105 metric tons of scintillator based on white-spirit $C_n H_{2n}$ ($\overline{n} \approx 9.6$) [6] and is situated in a salt mine at the depth of 570 m w.e. The natural radioactivity in the salt mine is ~ 300 times lower than in an ordinary environment [4]. One of the distinctive features of the ASD is its ability to detect about 70% of the energy of a shower coming from the rock.

Low natural radioactivity makes it possible to detect thermal neutrons using the reaction of their radiative capture by the scintillator hydrogen

$$\mathrm{n} + \mathrm{p} \to \mathrm{d} + \gamma . \tag{5}$$

A neutron, produced in the scintillator or coming from outside, slows down to thermal energies in collisions with carbon and hydrogen nuclei and after diffusion it is captured by a nucleus of hydrogen forming an excited nucleus of deuterium, which decays to the ground-state radiating γ-quantum with energy of 2.23 MeV. The average lifetime of the neutron with respect to the capture in the scintillator is about 180 μs. The ASD has detection efficiency of neutrons of $84 \pm 3\%$ for neutron sources uniformly distributed throughout the detector volume.

Pions with energies below 300 MeV slow down rapidly because of the ionization losses, 1–3% of them decaying on flight. Survived slow pions can reach a carbon nucleus and be captured to its K-shell. For negative particles the capture by a nucleus even from remote orbits is more probable than their decay.

Positive pions escape the capture and could be detected by the decay chain

$$\pi^+ \to \mu^+ + \nu_\mu , \tag{6}$$

$$\mu^+ \to \mathrm{e}^+ + \bar{\nu}_\mu + \nu_\mathrm{e} \qquad (\tau = 2.2 \cdot 10^{-6}\,\mathrm{s}) . \tag{7}$$

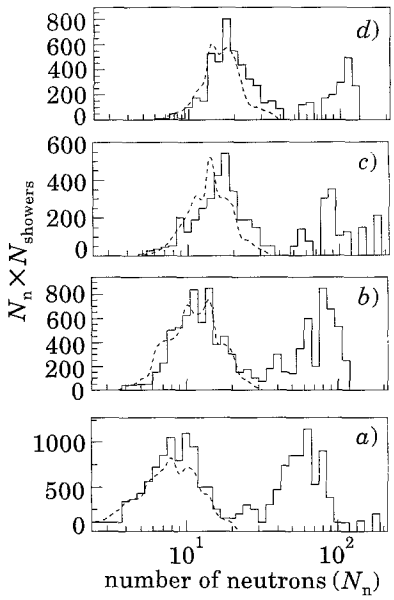


Fig. 1. – Dependence of the number of showers on the number of neutrons detected in them. The histograms are the experimental data, the curves Monte Carlo calculations for the following energy ranges: *a*) 90–115 GeV; *b*) 145–178 GeV; *c*) 212–250 GeV; *d*) 250–344 GeV.

The maximum positron energy is 53 MeV, the bulk of positrons having energies 30-50 MeV. However, after arrival of a big shower the detection system will be overloaded, and due to the dead time of $\tau_d \sim 5$ μs, the π^+ detection efficiency in the ASD is low, $\eta_\pi \leqslant 6\%$.

The high efficiency of neutron detection and the calorimetric nature of the detector provides the possibility of studying not only the dependence of the average number of neutrons in the shower on the energy, but the distribution of the showers according to the number of neutrons as well. However, to do this, it is necessary to separate perfectly electromagnetic and nuclear showers generated by muons.

Figure 1 shows the experimental dependence of the number of showers on the number of neutrons detected in them for 4 energy intervals. One can readily distinguish two local maxima. The average number of neutrons $\overline{N}_n$ for the left maximum corresponds to the value calculated for electromagnetic showers, and the number for the right maximum corresponds to the value for nuclear showers. The left maximum is well described by Monte Carlo calculations for electromagnetic showers. As follows from fig. 1, nuclear and electromagnetic showers are well separated by the number of neutrons in them.

The experimental data on the number of detected neutrons were then converted into the number of neutrons produced in the detector taking into account the detection efficiency and dead time. The dependence of the average number of neutrons on the shower energy release is shown in fig. 2. Curves 1 and 2 are calculations for electromagnetic and nuclear showers, respectively.

In electromagnetic showers neutron cascade curves practically coincide with the electron-photon ones. The calculations were made for several shower energies. The distribution of neutrons over the length of a nuclear shower developed in

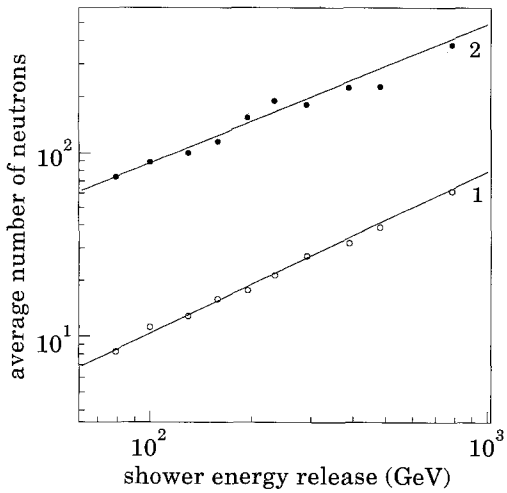


Fig. 2. – Dependence of the average number of neutrons on the energy release of the shower in the detector. ○ and ● represent the experimental data for electromagnetic and nuclear showers, respectively, corrected for the missed counts and detector efficiency. The straight lines 1 and 2 are the results of calculations.

a light material turns out to be similar in shape to the electron-photon cascade. Consequently the measured fraction of neutrons in showers of the two types is close to the measured fraction of shower energy. The average number of neutrons in nuclear ($\overline{N}_n^{nucl}$) and electromagnetic ($\overline{N}_n^{e.m.}$) showers can be represented as functions of the shower energy release E and the effective energy E_μ of the muon, which

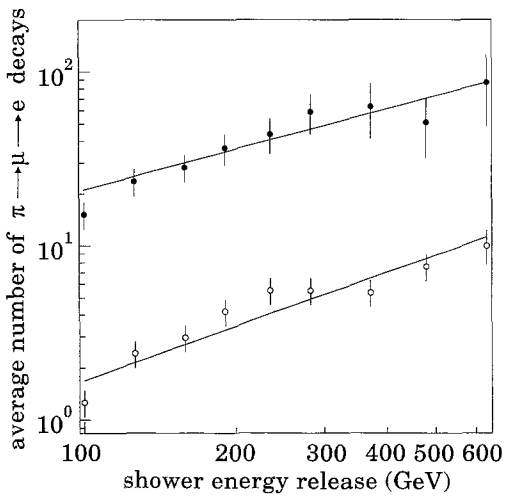


Fig. 3. – Average number of $\pi \rightarrow \mu \rightarrow e$ decays in electromagnetic (○) and nuclear (●) showers as functions of the shower energy release in the detector. The straight lines are power law approximations.

initiates this shower

$$(8)\qquad \begin{cases} \overline{N}_{\mathrm{n}}^{\mathrm{e.m.}} = k_1 E^{0.92 \pm 0.04}, & \overline{N}_{\mathrm{n}}^{\mathrm{e.m.}} = k_2 E_{\mu}^{1.03 \pm 0.05}, \\ \overline{N}_{\mathrm{n}}^{\mathrm{nucl}} = k_3 E^{0.69 \pm 0.04}, & \overline{N}_{\mathrm{n}}^{\mathrm{nucl}} = k_4 E_{\mu}^{0.78 \pm 0.05}. \end{cases}$$

In order to obtain the dependence of the average number of pions in nuclear and electromagnetic showers on the shower energy release, the showers were first separated according to the number of neutrons in them. The experimentally determined average number of $\pi \to \mu \to \mathrm{e}$ decays for electromagnetic and nuclear showers ($\overline{N}_{\pi}^{\mathrm{e.m.}}$ and $\overline{N}_{\pi}^{\mathrm{nucl}}$, respectively), with allowance for the detection efficiency, as well as the approximation curves, are shown in fig. 3. For electromagnetic showers the dependence on the shower energy release is well approximated by the formula

$$(9)\qquad \overline{N}_{\pi}^{\mathrm{e.m.}} \propto E^{1.0 \pm 0.15},$$

and for nuclear showers

$$(10)\qquad \overline{N}_{\pi}^{\mathrm{nucl}} \propto E^{0.8 \pm 0.2}.$$

The error in the experimental determination of the average number of pions includes the statistical errors in the number of showers and the number of pions in them, as well as the error due to inexact knowledge of the energy-dependent detection efficiency, and amounts to about 10–30%.

4. – Measurements of the neutron energy spectrum underground.

Flux and energy spectrum of neutrons coming from the rock with the kinetic energies T_{n} of 18–92 MeV were measured at the ASD [7] in addition to studying the dependence of the total number of neutrons on the energy transfer in the interaction.

The high-energy neutron flux was found in investigations of the delayed coincidence between two pulses produced by a neutron with $T_{\mathrm{n}} > 18$ MeV. The first pulse corresponds to the recoil protons and neutron energy losses due to inelastic interactions with carbon nuclei. The second one is due to the radiative capture (5) of the thermalized neutron itself. For the measurements the amplitude range of leading pulses $A = (10\text{–}50)$ MeV was selected, which corresponds to $T_{\mathrm{n}} = (18\text{–}92)$ MeV. The correspondence between T_{n} and A was obtained in Monte Carlo calculations of the detector response. The moments of arrival of the delayed pulses were recorded in the time gate of $\Delta t = (10\text{–}700)$ μs. The time distribution for interactions (5) is exponential, $\exp[-t/\tau]$, $\tau \approx 180$ μs, while uncorrelated background pulses are distributed uniformly in the time gate Δt. The neutron flux was determined by evaluating the exponential admixture in the time distribution with the use of a best-fit procedure, as well as by direct subtraction of the uncorrelated background. In the latter case, artificial triggering of the detector by a pulse generator was performed. Accounting for the detection efficiency, the flux of neutrons with kinetic energies $T_{\mathrm{n}} = (18\text{–}92)$ MeV is $2.3 \cdot 10^{-4}\,\mathrm{m}^{-2}\,\mathrm{s}^{-1}$.

Then, the whole range of leading pulse amplitudes A was divided into 8 bins of $\Delta A = 5$ MeV. For each bin the time and multiplicity distributions of trailing pulses were constructed, and the above-described procedure of neutron flux determination was repeated. Figure 4 demonstrates the so-obtained neutron energy spectrum [8]

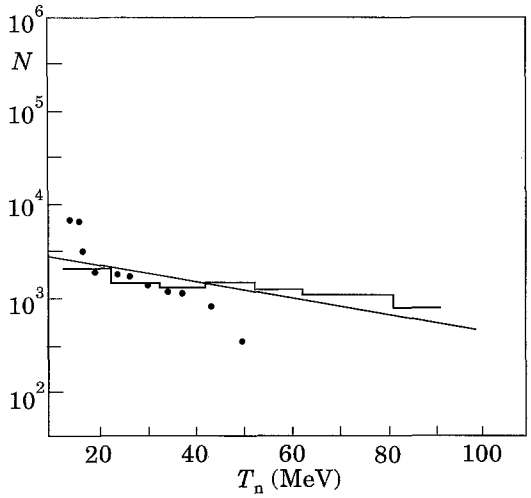


Fig. 4.

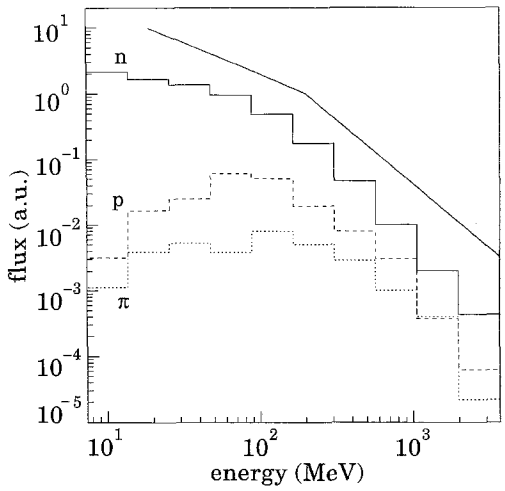


Fig. 5.

Fig. 4. – Energy spectrum of isolated neutrons underground. Histogram: our data; ●: data from [9], curve: results of Monte Carlo calculations.

Fig. 5. – Energy spectrum of neutrons passed through the shield of 250 g/cm^2 together with the spectra of secondary protons and pions at the same distance (solid, dashed and dotted histograms, respectively). The curve is the energy spectrum of the neutrons before the shield.

(the histogram), which is compared with the Monte-Carlo calculations and with the spectrum obtained by Barton at the depth of 60 m w.e. [9]. Both experimental spectra agree by shape in the energy range of $T_n = (20\text{–}40)$ MeV. At $T_n < 20$ MeV the admixture of neutrons produced in μ^- captures by nuclei could be seen in Barton's spectrum. This process is essential at low depths [1, 10]. For $T_n > 40$ MeV the detection efficiency in the experiment [9] falls down.

Our neutron energy spectrum could be approximated as

$$F_n(T_n) = cT_n^{-0.5 \pm 0.1}. \tag{11}$$

Calculations of the energy spectra of neutrons generated in showers and passed through a shield of a different thickness were presented in [11]. If the energy distribution of incident neutrons is taken as

$$F_n(T_n)\,dT_n \propto \frac{dT_n}{T_n}, \qquad (T_n \leqslant 200 \text{ MeV}),$$

$$F_n(T_n)\,dT_n \propto \frac{dT_n}{T_n^2}, \qquad (T_n > 200 \text{ MeV})$$

(the spectrum of neutrons generated in nuclear showers), then the spectrum of neutrons passed through the shield of 250 g/cm^2 thickness looks like that in fig. 5. The energy spectra of secondary pions and protons produced in the shield by the neutrons are shown in fig. 5, too. Behind the shield, neutron contribution is about 91%, proton

7% and pion 2% of the total particle flux. The neutron spectrum shown in fig. 5 has the same shape in the range of $T_n = (20\text{–}100)$ MeV as (11). So, after the passage through matter the spectrum dT_n/T_n ($T_n < 200$ MeV) changes to $dT_n/T_n^{0.5}$, due to the dependence of the neutron interaction cross-section on energy.

5. – The depth-neutron intensity curve.

It has been shown earlier [1,3] that the intensity of muon-induced hadrons falls down with depth slower than the muon intensity. Due to cascade processes, it depends on the average muon energy at the given depth as $\overline{E}_\mu^{0.7}$.

There are four main processes of neutron production by muons underground:

– μ^- capture by a nucleus followed by neutron emission; this process is important at depths less than 300 m w.e.;

– neutron photoproduction in muon inelastic scattering on nuclei;

– neutron production by hadrons in the muon-generated nuclear showers;

– neutron production by γ's in the muon-induced electromagnetic showers.

Figure 6 shows the dependence of the contribution of neutrons produced in the muon inelastic interactions on the interaction energy transfer E_t (MeV) for different depths. The first local maximum corresponds to an excitation of a giant resonance in nuclei of matter by photonuclear muon interaction with low E_t's. The second

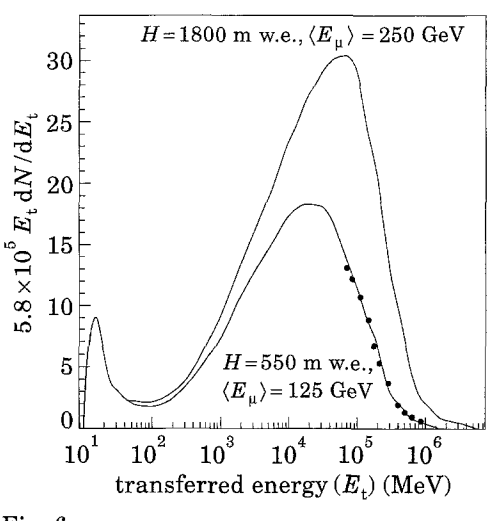


Fig. 6.

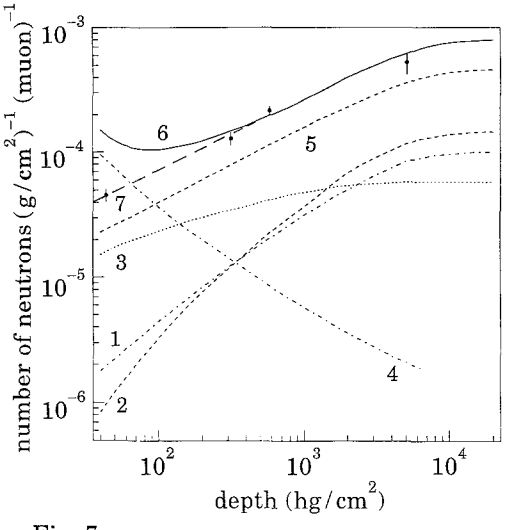


Fig. 7.

Fig. 6. – Dependence of the contribution of neutrons, produced in inelastic muon interactions, on the energy transfer in the interaction E_t. Points are the data from [7].

Fig. 7. – The number of generated neutrons per 1 muon per 1 g/cm^2 *vs.* the depth from the top of the atmosphere. Curves 1-3 correspond to the neutron production in electromagnetic showers, initiated by: muon bremsstrahlung (1), pair production (2), δ-electrons (3). Curve 4: μ^- capture. Curve 5: neutron production in inelastic muon interactions with the account for nuclear showers. Curve 6: the sum of all processes. Curve 7: all processes excluding μ^- capture.

maximum appears because of the multiple neutron production in the muon-generated nuclear showers, when the energy transfer $E_t > 1$ GeV. The total number of neutrons under the first local maximum can be comparable with that for the second one only for depths of < 50 m w.e. At bigger depths the hadrons are produced mainly in the nuclear showers.

The dependence of the number of generated neutrons per 1 muon per 1 g/cm^2 on the depth is shown in fig. 7. The curves 1-3 correspond to the neutron production in electromagnetic showers initiated by muon bremsstrahlung (1), by muon-produced $e^+ e^-$ and δ-electrons (2 and 3), respectively. Curve 4 corresponds to μ^- capture, and curve 5 to the neutron production in inelastic muon interactions with the account for nuclear showers. Curve 6 is the sum of all processes. The points represent experimental data for the depths of 25, 316, 570 and 5200 m w.e. The first 3 points were obtained at the Arteomovsk Scientific Station of the INR (Moscow) [3, 4, 7] and the fourth one at the Mont Blanc Laboratory in collaboration with the group of prof. C. Castagnoli [5]. All these experiments were performed using liquid-scintillation counters. At the depths of 25 and 316 m w.e. only events with the through-going muons were selected to exclude neutrons produced in the captures of stopping negative muons. The calculations and experimental data agree quite well.

The total number of neutrons produced by 1 muon in 1 g/cm^2, excluding the well-known process of muon capture, depends on the average muon energy at a given depth as

$$N_n(H) = N_0 \bar{E}_\mu(H)^{0.75 \pm 0.05} . \tag{12}$$

If $N_n(H_0)$ is known for some depth H_0, then for another depth H

$$N_n(H) = \frac{\mathscr{I}_\mu(H)}{\mathscr{I}_\mu(H_0)} \left[\frac{\bar{E}_\mu(H)}{\bar{E}_\mu(H_0)} \right]^{0.75} N_n(H_0), \tag{13}$$

where $\mathscr{I}_\mu$ is the vertical muon intensity. This method of scaling the number of generated neutrons practically does not depend on the interaction model.

6. – Electromagnetic component produced by cosmic-ray muons underground.

In passing through the matter cosmic-ray muons lose their energy due to production of knock-on electrons, $e^+ e^-$-pairs, bremsstrahlung and inelastic scattering. The electrons and gammas produced in these processes initiate electromagnetic showers, while $\pi^\pm$ from muon inelastic-scattering nuclear showers. The secondary particles in small electromagnetic showers constitute the large part of the background in the energy range of 5–200 MeV.

The secondary electromagnetic component could disturb the spatial resolution of the detectors with counter systems, designed for measurements of high-energy muon tracks. This background is important especially if a detector consists of light elements and cascade particles can go far away from the muon track.

Muon energy spectrum underground can be approximated by the following expression [12]:

$$(14)\qquad \begin{cases} \dfrac{\partial N_\mu(E, H)}{\partial E} = A(H)[\Delta E(H) + E]^{-\gamma+1}, & \text{where} \\ \Delta E(H) = \dfrac{\{1 - \exp[-(b_p + (b_\gamma + b_n)k)H]\}\beta}{b_p + (b_\gamma + b_n)k}, \end{cases}$$

b_p, b_γ, b_n are the mean relative energy losses due to pair production, bremsstrahlung and inelastic scattering, respectively,

$$b_{p,\gamma,n} = \int_0^1 v \frac{\mathrm{d}\sigma}{\mathrm{d}v}\,\mathrm{d}v,$$

v is the fraction of energy lost by muon, β the energy loss due to ionization, $k = k(\gamma, b_\gamma, b_n) < 1$ and takes into account fluctuations of the energy losses due to bremsstrahlung and inelastic scattering.

As the number of nuclear showers underground is several times lower than the number of electromagnetic ones [13], the main part of secondary electromagnetic particles is produced in the electromagnetic showers. So one can consider three processes: bremsstrahlung, production of knock-on electrons and pair production.

The equilibrium energy spectrum of cascade particles produced by muons in the i-th process, normalized per one muon, has been obtained in [14] for different depths

$$(15)\qquad \frac{\partial N(E_{\mathrm{cp}}, H)}{\partial E_{\mathrm{cp}}} = \frac{\displaystyle\int_{E_{\mathrm{cp}}}^{\infty} \frac{\partial N_{\mathrm{cp}}(E_{\mathrm{cp}}, E_i)}{\partial E_{\mathrm{cp}}}\,\mathrm{d}E_i \int_{E_i}^{\infty} \frac{\partial N_\mu(E_\mu, H)}{\partial E_\mu} \frac{\partial \sigma_i(E_\mu, E_i)}{\partial E_i}\,\mathrm{d}E_\mu}{\displaystyle\int_0^{\infty} \frac{\partial N_\mu(E_\mu, H)}{\partial E_\mu}\,\mathrm{d}E_\mu},$$

where i denotes the knock-on electrons, electron-positron pairs or gammas from bremsstrahlung, $\partial N_{\mathrm{cp}}(E_{\mathrm{cp}}, E_i)/\partial E_{\mathrm{cp}}$ is the differential spectrum of cascade particles going in equilibrium with the i-th parent, $\partial N_\mu(E_\mu, H)/\partial E_\mu$ the spectrum of muons at a given depth H, $\partial\sigma(E_\mu, E_i)/\partial E_i$ the cross-section of the corresponding process.

The number of secondary particles depends on the depth as

$$(16)\qquad N(H) = \sum_i \int_0^{\infty} \frac{\partial N(E_{\mathrm{cp}}, H)}{\partial E_{\mathrm{cp}}}\,\mathrm{d}E_{\mathrm{cp}}.$$

Figure 8 shows the results for the secondary γ's. It is very interesting that for the depths of $H > 3000$ m w.e. the main process is the pair production and the average number of the equilibrium gammas is more than 1. It means that any muon is accompanied by a γ-quantum with the average energy of $\sim 1/3$ of the critical one [2]. Some γ's decline on rather big angles from the track of the parent muon and can come into the detector like isolated particles, producing background events.

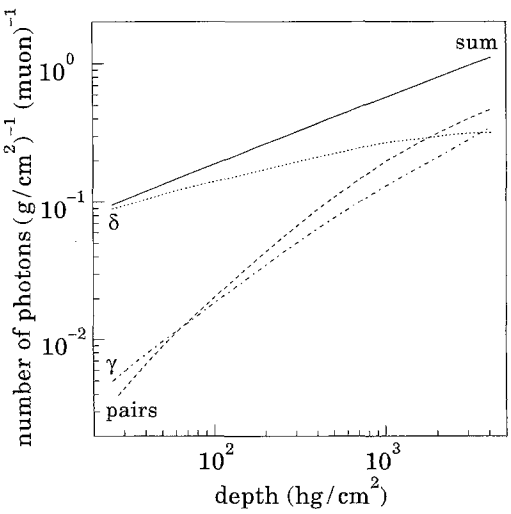


Fig. 8. – Average number of secondary γ's per one muon *vs.* the depth. Symbols near the curves indicate cascade parents: γ: γ-quanta, δ: δ-electrons, pairs: e^+e^--pairs, sum: the sum of all processes.

7. – Isotope production underground.

Nuclear and electromagnetic showers, passing through detectors, produce not only nuclear-active particles, but also labile nuclides in πA-, γA- or nA-interactions. Reactions of this type can cause background, which is difficult to get rid of, when long-lived isotopes decaying with the emission of high-energy β's or γ's are produced.

(n, p)-reactions with ^{16}O or ^{12}C could be the examples of background creation in water Cherenkov and scintillation counters:

$$(17)\qquad \begin{cases} {}^{16}O + n \to {}^{16}N + p\,, \\ {}^{16}N \to {}^{16}O + e^- + \bar{\nu}_e\,, \end{cases} \qquad \begin{cases} \tau_{1/2} = 7.14\ \text{s}\,, \quad E_e^{\max} = 10.44\ \text{MeV}\,(26\%)\,, \\ E_e^{\max} = 4.3\ \text{MeV}\,, \quad E_\gamma = 6.13\ \text{MeV}\,(68\%)\,, \\ E_e^{\max} = 3.3\ \text{MeV}\,, \quad E_\gamma = 7.13\ \text{MeV}\,(6\%)\,, \end{cases}$$

$$(18)\qquad \begin{cases} {}^{12}C + n \to {}^{12}B + p\,, \\ {}^{12}B \to {}^{12}C + e^- + \bar{\nu}_e\,, \end{cases} \qquad \tau_{1/2} = 0.03\ \text{s}\,, \quad E_e^{\max} = 13.37\ \text{MeV}\,(98.3\%)\,.$$

The former creates background for solar-neutrino measurements [15], the latter imitates the charged-current $\bar{\nu}_e$-reaction with carbon,

$$(19)\qquad {}^{12}C + \bar{\nu}_e \to {}^{12}B + e^+\,,$$

which is used to determine the upper limit on the $\bar{\nu}_e$ flux from relic supernovae [16] and limits on the parameters of flavour oscillations $\bar{\nu}_{\mu,\tau} \to \bar{\nu}_e$ in the supernova neutrino fluxes [17]. The previously obtained limits could be essentially improved after determining the background from interactions (18).

For scintillators containing fluorine, (n, γ)-reactions simulate detection of ν_e:

$$(20)\quad \begin{cases} {}^{19}\mathrm{F} + \mathrm{n} \to {}^{20}\mathrm{F} + \gamma\,, \\ {}^{20}\mathrm{F} \to {}^{20}\mathrm{Ne} + \mathrm{e}^- + \bar{\nu}_e\,, \end{cases} \qquad \tau_{1/2} = 12\ \mathrm{s}\,,\ E_e^{\max} = 6.7\ \mathrm{MeV}\,,\ E_\gamma = 6.83\ \mathrm{MeV}\,.$$

To detect low-energy neutrinos by carbon-containing scintillation detectors, knowledge of the ^{10}C, ^{11}C, ^{8}B, ^{8}Li, ^{11}Be production is essential. Some of these nuclides are produced together with neutrons, some of them are produced by neutrons, *e.g.* in reactions $^{12}\mathrm{C}(\gamma, \mathrm{n})^{11}\mathrm{C}$, $^{12}\mathrm{C}(\mathrm{n}, 2\mathrm{n})^{11}\mathrm{C}$, $^{12}\mathrm{C}(\mathrm{p}, 3\mathrm{n}2\mathrm{p})^{8}\mathrm{B}$, $^{12}\mathrm{C}(\pi^+, \alpha\pi^0)^{8}\mathrm{B}$, etc. [18]. In this case the nuclide production rate can be readily evaluated when knowing the total number of neutrons produced by muons, N_n^{tot}, and cross-sections of the (γ, n), (n, γ), (n, xn), (n, pxn), (n, dxn), (π, Xn), ... -reactions with the neutron participation ($x = 1, 2, 3, \ldots$). Our estimates give the total number of ^{11}C, ^{8}B + ^{8}Li, ^{12}B of ~ 20%, 5%, 1% of N_n^{tot}, respectively, for detectors with the scintillator C_nH_{2n}, and the total number of ^{16}N 3–3.5% of N_n^{tot}, for water Cherenkov detectors (reaction (17)).

Let us note that the dependence of the nuclide production rate on the depth is the same as for neutrons (see (12))

$$(21)\qquad N_{\mathrm{isot}}(H) \propto \bar{E}_\mu(H)^{0.75}\,,$$

so measurements of N_{isot} at some particular depth could help estimate the background at any depth. On the other hand, this experiment will give additional possibility to measure high-energy neutron flux. We are planning to use for this purpose the ASD in Arteomovsk ($H = 570$ m w.e.), where satisfactory statistics could be gained in quite a short time.

8. – Conclusions.

The results presented here show that an extensive both theoretical and experimental work involving the muon interactions in the matter has been done. It shows that the knowledge of production of secondary particles by cosmic-ray muons plays an important role for the evaluation of the background in experiments currently running in underground laboratories. This could be an utmost and decisive role for any quantitative expectation mainly in experiments searching for rare events.

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